

Roll No.....

Mid Semester Examination 2024

Name of the Program: **B.Tech.**

Semester: **VIII**

Name of the Course: **Mobile Ad Hoc Networks**

Course Code: **TEC 857**

Time: **90 minutes**

Maximum Marks: **50**

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries **10 marks**.

Q1	(10 marks)	CO1
(a)	Explain the basic operating principle of wireless ad hoc networks	
OR		
(b)	Point at least ten differences between cellular networks and ad hoc wireless networks.	
Q2	(10 marks)	CO2
(a)	Explain all the modifications incorporated by MACAW in back-off mechanism. Discuss with example.	
OR		
(b)	Point the issues and design goals of MAC protocols.	
Q3	(10 marks)	CO1
(a)	Explain the following i) IrDA ii) HomeRF	
OR		
(b)	Discuss the protocol stack of IEEE 802.16 standard. Point the differences between IEEE 802.11 and IEEE 802.16.	
Q4	(10 marks)	CO1/ CO2
(a)	How ad hoc networks is useful in military and emergency applications.	
OR		
(b)	Explain the “Hidden Terminal Problem” and “Exposed Terminal Problem” in ad hoc wireless networks.	
Q5	(10 marks)	CO1/CO2
(a)	Explain the “Hybrid Network” in MANET.	
OR		
(b)	How MACA (Multiple Access Collision Avoidance) protocol reduce the Hidden and Exposed Terminal Problems.	

